Fiscal Year:	FY 2009	**Task Last Updated:**	FY 09/15/2009
PI Name:	Brainard, George C. Ph.D.		
Project Title:	Optimizing Light for Long Duration Space Exploration		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19107-5083	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	09/01/2008	**End Date:**	08/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	7
No. of PhD Candidates:	0	**No. of Master' Degrees:**	5
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	11
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hanifin, John (Thomas Jefferson University) Pineda, Maria (Thomas Jefferson University) Lockley, Steven (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-HFP01605		
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Task Description:	The goal of this project is to determine the optimal lighting for use as a countermeasure to sleep and circadian disruption for astronauts. Sleep deficits are a risk factor during space missions. The resulting physiological and behavioral changes from circadian disruption can threaten the success of a mission by diminishing alertness, cognitive ability, and psychomotor performance. Currently, over 45% of all medications taken in space are sleep aids (Putcha et al., 1999). Despite the use of these, more than half of the astronauts still get only between 3 and 6 hours of sleep per night during space flight (Barger et al., 2009). Light is the primary stimulus for human circadian regulation. While on Earth, humans have a 24-hour day/night cycle to maintain healthy circadian entrainment. In space, astronauts must contend with either a rapidly changing or severely disrupted day/night cycle and must work in a dimly lit spacecraft interior with few or no windows. Bright white light has already been implemented as a pre-launch countermeasure but has yet to be used during space flight. Providing sufficient light intensities to work areas in the International Space Station (ISS), the planned Crew Exploration Vehicle (CEV), and future lunar habitats raises several concerns, among which are heat production, energy consumption, and upmass. Improving these factors requires a better understanding of how different artificial and natural ambient light sources regulate the human circadian system. From these data it may be possible to optimize astronaut and ground crew light exposure both as a stimulus for vision as well as a countermeasure for sleep and circadian disturbances before or during space exploration. Currently, NASA uses white fluorescent light for interior illumination of both the STS and the ISS at relatively low intensities, and for a pre-launch lighting countermeasure for circadian disruption in crew quarters at much higher intensities. In the previous funding period, our NSBRI research was supported to determine if those fluorescent lamps have increased efficacy for melatonin suppression and circadian phase-shifting when they are enriched in the blue portion of the spectrum. Philips Lighting, an NSBRI industry partner, developed and provided the prototype blue-enriched lamps and exposure systems used in that study. In addition, we began characterizing the circadian potency of ambient extraterrestrial light that astronauts will encounter on the surfaces of the Moon and Mars. To develop a lighting countermeasure for space exploration, it is vital to determine the sensitivity of astronauts' circadian systems to both ambient and artificial lighting stimuli. In the current period of research, we are shifting our efforts from testing fluorescent light sources to testing solid-state light sources. Artificial illumination for the CEV as well as the planned lunar and Martian habitats will be provided exclusively by light emitting diodes (LEDs) and other electroluminescent technologies. The following list identifies each specific aim and this past year’s progress. 1.a. Determine the neuroendocrine, alerting, and circadian potency of light emitted by the white LEDs being considered by NASA for CEV in-flight and lunar habitat lighting systems. Test the hypothesis that white LED light has increased efficacy for suppressing melatonin, phase-shifting, and enhancing alertness compared to the fluorescent lamps currently used by NASA. For this new aim, we worked with Apollo Health, a NSBRI industrial partner, to design, fabricate, and install two 4’ by 4’ white LED lighting systems in our laboratory. To date, the 5 healthy male and female subjects entered into the study have completed over 30 nighttime studies. 1.b. Assess the ocular safety of experimental subjects and astronauts exposed to light from the LEDs at intensity levels higher than those currently specified for visual stimulation in space habitats. Working from national and international safety guidelines, a medical physicist did a preliminary independent hazard analysis on the white light emitted by our two LED exposure systems to assure the ocular safety of our study subjects. A final safety analysis on this lighting system is now being written. 2. Determine the acute neuroendocrine, alerting, and performance enhancing potency of the ambient extraterrestrial light as filtered through spacesuit visors or glazed windows for Moon and Mars missions. Test the hypothesis that ambient lunar and Martian light, as filtered through spacesuit visors, will suppress melatonin less and will be weaker for enhancing alertness and performance compared to ambient daylight on Earth. For this new specific aim, additional optical elements have been developed and installed for simulating Earth daylight in a Ganzfeld dome. To date, 4 healthy subjects have entered into the Ganzfeld Earth daylight study and completed over 15 nighttime studies. In addition, continuing progress from the previous year, more than 20 new nighttime studies have been completed with simulated Moon and Mars light. Together, the research from the two prior funding cycles along with the studies performed in the past year provide significant progress towards the development of lighting countermeasures for sleep and circadian disruption in ground crew members and astronauts. The results will also impact the NASA Human Integration Design Handbook and the Space Flight Human Systems Standard, NASA-STD-3001 that provide guidance for crew health, habitability, environment, and human factors in human spaceflight. Our progress addresses Critical Risk areas 26, 27, and 44 in the Bioastronautics Roadmap (2005), and areas 20, 21, and 27 in the NASA Human Research Program Integrated Risk Plan (2009). These areas concern countermeasures that will optimally mitigate performance problems associated with sleep loss and circadian disturbances and the “mismatch between crew physical capabilities and task demands.”
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The knowledge we hope to gain from this research, though focused on spaceflight, will also benefit people on Earth. The sleep deficits experienced by astronauts during space flight can be considered a threat to the success of space missions (Bioastronautics Roadmap, Longnecker and Molins, 2005; NASA Human Research Program Integrated Risk Plan, 2009). The resulting physiological and behavioral changes caused by sleep and circadian disruption can lead to diminished alertness, cognitive ability and psychomotor performance (Dijk et al., 2001). As a measure to counteract sleep disruptions, over 45% of all medications taken in space are sleep aids (Putcha et al., 1999). A significant portion of the global population suffers from chronic sleep loss and/or circadian-related disorders. Evidence for disease or illness occurring due to a disruption of circadian homeostasis has mounted significantly in the past several years. In the United States, nearly 22 million Americans do shift work that interferes with a biologically healthy nocturnal sleep cycle (US Bureau of Labor Statistics, 2007). Shift workers have been shown to be more likely to suffer from a wide variety of ailments, including cardiovascular disease, gastrointestinal distress, and cognitive and emotional problems. Furthermore, epidemiological studies of female shift workers have shown that they are statistically more likely to suffer from breast and colon cancer as compared to their day shift counterparts. The World Health Organization has identified shift work as a probable risk for cancer (The International Agency for Research on Cancer, 2007). Our laboratory is involved in testing the hypothesis that night time exposure to light suppresses melatonin and contributes to cancer risk (Blask et al., 2005; Stevens et al., 2007). Aside from evidence of a breakdown in physical health, the effects of circadian disruption and sleep loss are known to have potentially dangerous behavioral effects. Mental fatigue, diminished alertness, loss of

Research Impact/Earth Benefits:	psychomotor coordination and decreased physical performance are all commonly found in individuals with sleep loss, sleep debt, or circadian misalignment. The impact of these dangers affects many industries, including transportation, manufacturing, communications, and medicine. It has long been a source of concern for the military and, more recently, homeland security. Many people also experience the same effects after air travel across several time zones. In the past, the U.S. Air Force has supported our laboratory to study the acute alerting effects of light (French et al., 1990; Brainard et al., 1996). Our current work for NIH has continued this effort (Lockley et al., 2006). Existing light therapy interventions stand to benefit from enhancing our understanding of how different wavelengths of the spectrum affect human circadian and neurobehavioral regulation. The result may be a more efficient intervention with increased potency and/or fewer side effects. One such disorder currently being treated with bright white light is Seasonal Affective Disorder (SAD), also known as winter depression. It is estimated that as many as 1 in 5 Americans suffer from SAD or its milder version, subsyndromal Seasonal Affective Disorder (sSAD) (Lam and Levitt, 1999). Similar bright white light interventions are also used to treat jetlag. Side effects from exposure to bright white light for these and other therapies include: hypomania, headache, vision problems, nausea, dizziness, and anxiety. Optimizing the light spectrum for specific affective and/or circadian-related disorders may deliver the same medical benefit with lower levels of light intensity, and potentially fewer side effects. Our group has completed Phase I testing of a lower intensity of light therapy with blue solid-state lighting for SAD patients (Glickman et al., 2006).
Task Progress:	A major goal of our work is to optimize lighting that supports vision and serves as an effective circadian countermeasure for astronauts and ground crew during space missions. As we approach the end of the first year of this funding cycle, progress has been made on several of the new specific aims of our project. One of our specific aims is to determine the neuroendocrine potency of light emitted by the white LEDs being considered by NASA for CEV in-flight and lunar habitat lighting systems. Although illumination from the LEDs appears to be white to the visual system, it is significantly richer in blue wavelengths compared to the white fluorescent light currently used by NASA. We are testing the hypothesis that white LED light has increased efficacy for melatonin suppression compared to the white fluorescent light. Working with Apollo Health, an NSBRI industrial partner, we designed, fabricated, and installed two 4’ by 4’ white LED lighting systems in our laboratory. To date, the 5 healthy male and female subjects entered into the study have completed over 30 nighttime studies. Another specific aim is to assess the ocular safety of experimental subjects and astronauts exposed to light from the LEDs at intensity levels higher than those currently specified for visual stimulation in space vehicles and habitats. Working from national and international safety guidelines, a medical physicist has done a preliminary independent hazard analysis on the light emitted by our two experimental white LED exposure systems. A final hazard analysis is now being written. Since circadian and sleep regulation depends on an astronaut’s total exposure to both artificial light and passive ambient illumination, a third specific aim is to compare the potency of the ambient light on the surface of Earth, Moon, and Mars for acute melatonin suppression. In the last funding period, we worked with optical engineers to create a Ganzfeld lighting system to simulate extraterrestrial light that astronauts would see on the surface of Moon and Mars. This equipment then was used to begin testing the potency of simulated Moon and Mars light for its capacity to suppress melatonin in healthy men and women. In this past year, more than 20 new nighttime studies have been completed with simulated Moon and Mars light. Further, new optical elements have been installed for simulating Earth daylight in a Ganzfeld dome. To date, the 4 healthy subjects entered into the Ganzfeld Earth daylight study have completed over 15 nighttime studies. Finally, last year, we completed twenty 7-day in-patient circadian phase-shifting studies with healthy subjects comparing the potency of white-appearing, blue enriched fluorescent light (17,000 K) versus the white-appearing fluorescent light (4,000 K) currently used by NASA as a pre-launch countermeasure for astronauts. This year, continuing work involves analysis of neuroendocrine, circadian and neurobehavioral data from those studies.
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